

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	
Project Name	AI and Machine Learning Project
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	All Python, HTML, and CSS files follow consistent formatting.
2	Proper File Structure	Files and folders are logically organized	Yes	Project is organized into templates, static, models, data, and application files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as age, income, credit_score, and prediction_result improve readability.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like predict(), home(), and model-loading functions clearly indicate their purpose.
5	Code Comments	Inline and block comments explain logic	Yes	Comments are added for preprocessing, model loading, prediction, and routing.
6	Modular Design	Code is split into reusable functions/modules	Yes	Training script, Flask application, HTML

				templates, CSS, and ML model files are separated.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Only necessary functions and files are included in the final project.
8	Error Handling	Exceptions and errors are handled gracefully	Cefully Yes	Input validation and exception handling are implemented to avoid runtime errors.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	app.py	Python, Flask	Handles routing and prediction requests	High

2	train_model.py	Python, Scikit-learn	Model training and preprocessing	High
3	model.pkl	Joblib	Loaded for prediction without retraining	High
4	index.html	HTML5	User input interface	High
5	style.css	CSS3	Common styling across web pages	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate folders for templates, static files, data, and models.
Readability	5	Code uses meaningful variable names, proper formatting, and comments for better understanding.
Reusability	5	Modular design allows components such as the ML model and Flask application to be reused easily.
Documentation / Comments	5	Sufficient comments and README documentation make the project easy to understand and maintain.
Overall Score	5/5	The project follows good coding standards, modular architecture, and maintainable design practices.